

Z-Spend

On-chain-enforced agentic treasury — an AI agent that cannot overspend, even with its own key.

Tempo Track · Crypto World's Fair · Live on Moderato testnet

THE PROBLEM

Agentic AI wallets hold real funds, but "trust the agent" fails. A stolen or misconfigured key can drain the treasury. DeFi multisigs don't scale to agent autonomy.

THE BUILD — 2 LEGS OF DEFENSE

1. Policy layer (off-chain, fail-closed)

- Per-category daily caps, per-recipient caps, max single payment, reserve floor
- Every spend = verdict computed from immutable JSONL ledger
- 7/7 unit tests passing

2. On-chain enforcement (Tempo native)

- Agent signs as an authorized **access key** with pathUSD limit + scope
- Protocol enforces tip-1011 scopes + spend limits at tx validation
- Compromise the key — still can't exceed the allowance

LIVE EVIDENCE (REAL TXS ON MODERATO)

Step	Result
Authorize agent key (\$100/day, scoped pathUSD.transfer)	On-chain authorizeKey tx
pay-001 ops \$80	Policy APPROVE → executed via access key
pay-002 marketing \$160	Policy REJECT (cap \$150)
pay-003 ops \$100 (policy OK, key exhausted)	Blocked by chain — revert 0x8a9e71ea
Remaining allowance read	\$19.99 / \$100 — fees included

WHY WE WIN

- **Native, not fake:** uses Tempo's real primitives (TIP-20 pathUSD, account keychain, access-key scopes) — protocol-level understanding, not a port.
- **Working product on testnet:** full demo video + transcript + explorer-verifiable tx hashes.
- **Vertical fit:** Tempo = payments L1; agent treasuries are the #1 custody question it must answer.